

Probing Protected Attributes within General-Purpose LLMs for Hiring

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General-purpose Large Language Models (LLM) applications in human resource workflows are primarily driven by the need for objectivity and efficiency. However, a significant gap exists between enterprise-level AI and the consumer-grade "black-box" services frequently utilized by small and medium-sized enterprises (SMEs), raising critical questions regarding algorithmic consistency and fairness. This paper utilizes the *IN-OUT framework*, a stimulus-response methodology derived from the cognitive sciences, to empirically audit the behavior of one popular LLM. We employ a 2×2 factorial design to analyze how the model responds to protected attributes (including gender, age, and nationality) across five varying stimulus volumes ranging from $K = 16$ to 12,000 curricula vitae (CVs).

The findings reveal that while the model demonstrates acute sensitivity to objective professional suitability demographic bias does not operate as a static constant. Instead, bias manifests non-linearly as a context-dependent interaction effect. While the model exhibited robust neutrality at extreme low and high stimulus volumes ($K = 16$ and $K = 12,000$) significant gender and age-related structural breakdowns occurred at mid-range context densities ($K = 3,000$ and $K = 6,000$). In contrast, evaluations regarding nationality remained robustly equitable across all stimulus volumes.

Furthermore, a multi-account audit across $N = 11$ independent user sessions confirms that while stochastic noise significantly shifts absolute scoring for candidate filtering baselines, the underlying discriminatory configurations remain structurally invariant ($r > 0.99$) even at the smallest sample sizes. These results suggest that absolute score thresholds for candidate filtering are highly unreliable and that latent biases can trigger unpredictably as context windows approach saturation limits. The study concludes with practical implications for digital governance and the necessity of high-volume algorithmic stress tests to ensure non-discriminatory HR selection under the regulatory frameworks of the EU AI Act.

Keywords: Behaviorism, machine psychology, algorithmic management, bias, AI

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1 Introduction

The integration of general-purpose Large Language Models (LLMs) into human resource (HR) workflows is primarily driven by the dual objectives of automating labor-intensive manual tasks and enhancing the perceived

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quality of decision-making processes [1]. A prevalent assumption among many organizations is that LLMs operate neutrally, evaluating candidates based solely on objective information found in CVs, cover letters, and work samples [13]. In the domain of human resources, companies increasingly deploy these models to screen batches of application documents, where AI systems perform a preliminary selection that determines which candidates are excluded from further consideration [3]. CVs, a tabular summary of an applicant's professional development, still play a vital role in HR processes. Generic motivation letters, by contrast, are becoming less important, partly due to the widespread use of LLMs. This allows in-person interview time to be focused more effectively on the most adequate candidates [9]. However, this shift raises critical questions about the quality of these selections and the potential risk of applicants being discriminated against based on their protected attributes. [6].

A significant gap exists between enterprise-level AI solutions and the tools utilized by smaller organizations. Small and medium-sized enterprises (SMEs) with limited HR and AI budgets typically do not procure specialized, high-cost algorithmic hiring services [12]. Instead, they frequently rely on the "Swiss Army Knife" of general-purpose consumer services, such as ChatGPT, Gemini, Claude, and others to manage recruiting processes. This reliance introduces significant questions regarding data protection, consistency, and the fairness of these "black-box" systems. To address the challenge of studying black-box models, we adopt a stimulus-response methodology derived from the cognitive sciences, operationalized through the IN-OUT framework [5]. The core principle involves the systematic manipulation of stimulus material, comprising prompts and attached files, to produce an output that reveals the hidden behavioral patterns of the model.

Ensuring scientific reproducibility in auditing consumer-grade services is challenged by non-API access, restricted processing, and unknown, frequently updated model architectures [7]. Our research seeks to empirically audit the behavior of general-purpose LLMs in light of European Union legislative efforts, including the regulatory frameworks outlined in the EU AI Act. Our core hypothesis is that, when faced with thousands of CVs, the probabilistic noise of individual CVs diminishes, potentially revealing underlying systematic biases. [14].

To test this, we utilize the IN-OUT framework [5] to analyze model responses to protected attributes, specifically gender, age, and nationality, across five varying stimulus volumes (K) ranging from 16 to 12,000 CVs. We aim to determine if variability observed in large-scale batches is present at the smaller scales typical of SME application pools. The stimulus material was procedurally generated using a randomized Python workflow to ensure distributional variety while controlling for professional suitability.

While LLMs are often evaluated through single-turn deterministic outputs, such snapshots may fail to capture the full distributional profile of the model's internal associations. Although users cannot control the "temperature" parameter in consumer interfaces, repeating standardized queries allows underlying statistical biases and societal stereotypes to manifest as measurable patterns [10]. In this context, "suitability scores" serve as collective evidence of systematic preferences, bias, or "machine psychology." Consequently, this repeated sampling acts as a stress test for fairness, surfacing latent biases that remain obscured in one-shot settings.

For this experiment, CVs were classified as "suitable" (industry experience) or "unsuitable" (no industry experience) for a postal delivery position, and the model's sensitivity to this manipulation was verified. To examine the impact of instructional framing, we mandated role-adopting behavior by instructing the model to act as a professional Human Resource Manager. For the smallest stimulus set ($K = 16$) we reproduced findings across a cohort of $N = 11$ independent accounts to audit inter-user variation. This study provides first insights into the

risks of utilizing consumer-grade LLMs in HR pre-selection and contributes to developing practical standards for non-discrimination in digital innovation governance.

This study addresses three interconnected research questions. **Research question 1** examines whether the model accurately translates objective professional suitability into differentiated scores, validating the scoring mechanism as a reliable measurement instrument before investigating bias. **Research question 2** investigates whether the model’s evaluations are systematically influenced by the protected attributes of gender, age, and nationality, and whether any observed discrimination gaps vary as a function of stimulus volume. **Research question 3** analyzes the structural stability of the model’s evaluative pattern across increasing stimulus volumes, from $K = 16$ to $K = 12,000$ CVs, and across $N = 11$ independent user accounts, to distinguish inherent algorithmic bias from session-specific stochastic noise.

2 Method

This study utilizes the *IN-OUT framework*, a stimulus-response methodology derived from the cognitive sciences, to audit the behavior of LLM [5]. The core principle of this framework is the systematic manipulation of input stimuli (comprising specific prompts and attached files) to observe the resulting output and infer the model’s internal decision logic. By analyzing these responses, the framework identifies hidden patterns and potential biases that reflect the model’s underlying structure. This approach is particularly relevant for auditing general-purpose consumer services, where scientific reproducibility is often challenged by a lack of transparency regarding model weights, training data and the model structure. Consequently, the IN-OUT framework enables a rigorous empirical audit of algorithmic management tools without requiring direct access to the system’s proprietary internal components.

2.1 Model Card

Although this study was initially aimed to evaluate different LLMs, Gemini and Claude were ultimately excluded from the analysis. These systems consistently failed to complete the required experimental workflows and frequently aborted the processing of the excessive stimulus material. Potentially, the services require to subscribe to more powerful service levels. However, clear responses were not given by the services. In the future, these services will be again considered for testing. To ensure scientific rigor and a deeper technical audit of the model’s behavior, the investigation was consequently narrowed to a single environment. To ensure transparency and scientific reproducibility, the technical specifications of the investigated system (GPT-5.2 Thinking) are detailed in the Appendix.

2.2 Prompt: Job Advertisement and Instructional Framing (IN)

The experimental stimulus was anchored by a standardized job advertisement for a postal delivery position. The outlined responsibilities included the delivery of letters and parcels, averaging under 10 kilograms, utilizing company-provided assets, such as electric vehicles, across a flexible five-day work week spanning Monday to Saturday. Candidate requirements specified the possession of a valid driver’s license, proficiency in the German language, and physical resilience to diverse weather conditions. Furthermore, the professional profile emphasized qualitative attributes such as reliability, engagement, and a consistent work ethic.

To standardize the evaluation process, a specific instructional framing prompt was employed for the LLM. The LLM was instructed to adopt the persona of a professional Human Resource Manager and perform a holistic assessment of the CVs provided. Suitability was quantified through a numerical score ranging from 1, representing an entirely unsuitable candidate, to 20, representing a highly suitable candidate. To ensure the independence of each evaluation and mitigate potential position-order effects, the prompt mandated that the LLM process each CVs individually within its attention window and reset its internal context between assessments. Additionally, web-recherche capabilities were explicitly disabled to isolate the internal biases of the LLM. The output was requested as a comma-separated sequence of scores delivered in a text file to facilitate automated downstream analysis. The complete wording of the instructional prompt and the job advertisement utilized in this study is provided in the Appendix.

2.3 CV Generation

The stimulus material was procedurally generated using a custom Python framework, `generator.py` (see repository), designed to simulate a realistic pool of applicants for a postal delivery position. The generation process followed a configuration-driven structure, where experimental groups, representing specific combinations of protected attributes and professional suitability, were defined in separate configuration files.

For each document, the script constructed a multi-section CV including personal details, practical experience, and education by randomly sampling from these group-specific attribute lists. To operationalize professional suitability, candidates in the “suitable” groups were assigned professional milestones within the logistics sector, while “unsuitable” candidates were assigned experiences in unrelated industries.

To investigate the effects of stimulus volume on the decision-making patterns of the LLM, five distinct batch sizes ($K \in \{16, 1,500, 3,000, 6,000, 12,000\}$) were generated. Scaling was achieved by adjusting the CVs-per-group parameter and employing specific random seeds for each batch to ensure full scientific reproducibility. Upon generation, the CVs from all experimental groups were merged into a single PDF document. The final sequence was randomized using a seeded shuffle function to prevent position-order bias. The complete generation script, configuration files, and synthesized CV datasets are publicly available on the public repository Open Science Framework at <https://osf.io/6h5xm/>.

2.4 Experimental Design

The study utilizes a 2×2 factorial design to systematically investigate evaluative biases within the LLM. The primary factor consist of protected attributes, specifically gender (male vs. non-male), age (young vs. non-young), and nationality (German vs. non-German), crossed with the candidate’s professional suitability (suitable vs. unsuitable) as second factor (Table 1). Suitability was always operationalized through the presence or absence of relevant industry experience in the logistics sector. The capacity of the model to recognize the suitability of the candidate based on different levels of industry experience was tested beforehand 5.

To isolate the effect of the three protected attributes on the language model’s evaluation, all other CV content was held constant across groups. Specifically, the pools of last names (Müller, Schmidt, Schneider, Fischer, Weber, Meyer, Wagner, Schulz, Bauer, Becker), places of birth (ten major German cities), cities of residence, and street names were identical in all conditions, ensuring that the perceived geographic and cultural background could not

Table 1. Operationalization of the three protected attributes in the CV generator. For each attribute, an advantaged and a disadvantaged category was defined.

Attribute	Gender	Age	Nationality
Realized via	First name	Year of birth	Citizenship
Advantaged	<i>male</i> : Noah, Matteo, Leon, Luca, Paul, Henry, Theo, Emil, Elias, Louis	<i>young</i> (1990–1999): 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999	<i>German-speaking</i> : German, Austrian, Swiss
Disadvantaged	<i>non-male</i> : Sophia, Emilia, Emma, Mia, Lina, Hanna, Ella, Mila, Klara, Lia	<i>non-young</i> (1970–1979): 1970, 1971, 1972, 1973, 1974, 1975, 1976, 1977, 1978, 1979	<i>Non-German-speaking</i> : Danish, Dutch, Belgian, Czech

confound the nationality manipulation. Educational background was also kept constant, including the set of school degrees (general higher education entrance qualification, secondary school leaving certificate), school grades, training programs (Business Administration Bachelor, IT Specialist), and training grades. Finally, the pools for spoken languages and hobbies were held identical across all groups. From these identical pools, the CV generation algorithm sampled randomly for each individual CV, such that any systematic differences in the model’s output between conditions can be attributed to the manipulated protected attribute rather than to incidental variation in other CV features.

To observe the impact of stimulus volume on the decision-making patterns of the LLM, this factorial structure was replicated across five distinct volume tiers (K), ranging from individual samples to large-scale batches. This design allowed for the identification of potential shifts in the evaluation logic as the data volume increased. An overview of the experimental parameters and conditions is provided in Table 2.

For the protected attribute gender, the possibly advantaged attribute was male and the possibly disadvantaged attributed was female. This assumptions was based on the observation of data-trained models, which were influenced by historical observations. The suggested advantaged attribute for age was ‘young’, as stereotypes might not associate fitness or cognitive performance with increasing age, both of which can be important in logistics. The disadvantaged attribute was “non-young”. For the protected attribute nationality, the advantaged attribute was German, for language speaking capacity. The disadvantage attribute was “non-German”.

2.5 Data Analysis & Metrics

The evaluation of the outputs generated by the LLM was conducted using a multi-stage statistical approach to ensure both the validity of the scoring mechanism and the identification of systematic biases.

Table 2. Summary of Experimental Design Factors and Scaling

Design Element	Specifications
Factorial Structure	2×2 (Protected Attribute $\times$ Suitability)
Protected Attributes	Gender (male, non-male), Age (young, non-young), Nationality (german, non-german)
Suitability Levels	suitable (Industry Experience), unsuitable (No Experience)
Stimulus Volumes (K)	16, 1,500, 3,000, 6,000, 12,000
Scoring Metric	Suitability Score (1–20)

2.5.1 Research question 1: Validation of Suitability Scores

The sensitivity of the LLM to the professional suitability of candidates was verified to ensure that the assigned scores accurately reflected the manipulated industry experience. Descriptive statistics, including means, standard deviations, and medians, were calculated for each group tier. The statistical significance of the score distribution across suitability levels was assessed using the Kruskal-Wallis test. Furthermore, the monotonic relationship between candidate experience levels and the assigned suitability scores was quantified using the Spearman rank correlation (ρ) and Kendall’s τ . A linear trend model was also employed to confirm the increase in score points per group stage. To evaluate the model’s sensitivity to gradual increments in professional suitability, candidate experience was systematically manipulated across four distinct tiers (20% as least experience, 40%, 60%, and 80% as most experience), generating a dataset of 3,000 CVs per condition. The relationship between these incremental experience levels and the assigned suitability scores was subsequently quantified using Spearman’s rank correlation, Kendall’s tau, and a linear trend model.

2.5.2 Research question 2: Bias Detection and Statistics

To identify potential patterns of discrimination, we used both descriptive and inferential metrics:

- **Discrimination Gap Analysis:** A descriptive bias measure, the Discrimination Gap (Δ), was calculated as the difference between the mean scores of advantaged and disadvantaged candidates, segmented by suitability level and stimulus volume (K).
- **Statistical Modeling:** Systematic bias was tested using an Ordinary Least Squares (OLS) regression model with robust standard errors (heteroscedasticity). The model investigated the interaction effects between the protected attribute (gender), professional suitability, and stimulus volume ($Score \sim Gender \times Suitability \times K$).
- **Multiple Testing Correction:** Separate 2×2 tests for each stimulus volume were conducted, with p-values adjusted using the Holm-Bonferroni method to control for Type I errors across the various K conditions.

2.5.3 Research question 3: Pattern Stability Analysis

- **M3A Stimulus Set Scaling:** We examined how the underlying evaluative pattern, defined by the intersection of protected attributes and professional suitability, behaves as the stimulus volume increases. The stability of this 2×2 configuration was tested across five distinct scaling tiers ($K \in \{16, 1, 500, 3, 000, 6, 000, 12, 000\}$) by calculating the correlation between group-mean score vectors. This analysis identifies whether the relative biases and scoring gaps observed at a small scale remain invariant or undergo non-linear shifts when the model processes high-density data batches.

- **M3B Multi-Account Consistency:** To distinguish systemic algorithmic bias from session-specific stochasticity, the stability of the evaluation pattern was audited across 11 independent user accounts. By replicating the baseline experiment ($K = 16$) across these accounts, we quantified the "probabilistic noise" and "temperature parameter" inherent in the LLM and assessed whether the 2×2 scoring configuration remains consistent across different environments. This comparison provides a reliability benchmark for audits conducted in non-API, consumer-grade platform settings.

3 Results

3.1 Research question 1: Model Sensitivity: Suitability Score

3.1.1 Baseline Performance:

The sensitivity of the Large Language Model (LLM) to the professional suitability of candidates was verified to ensure that the assigned scores accurately reflected the manipulated industry experience. Across all pooled conditions, candidates with suitable profiles received substantially higher suitability scores compared to unsuitable candidates regardless of their demographic group. Specifically, advantaged suitable candidates ($M = 17.86$, $SD = 2.02$) and disadvantaged suitable candidates ($M = 17.65$, $SD = 2.02$) significantly outperformed their unsuitable counterparts (advantaged: $M = 8.22$, $SD = 2.43$; disadvantaged: $M = 8.39$, $SD = 2.60$).

The score distribution across suitability levels was confirmed to be statistically significant via a Kruskal-Wallis test ($H = 50581.09$, $p < 0.001$). Furthermore, a strong monotonic relationship between candidate experience levels and suitability scores was established as quantified by both Spearman's rank correlation ($\rho = 0.865$, $p < 0.001$) and Kendall's tau ($\tau = 0.738$, $p < 0.001$). A linear trend model corroborated this relationship indicating a significant increase of 9.45 score points per suitability level ($t = 537.96$, $p < 0.001$). As illustrated in Figure 1, suitability scores significantly differ between suitable and unsuitable candidates across all examined categories.

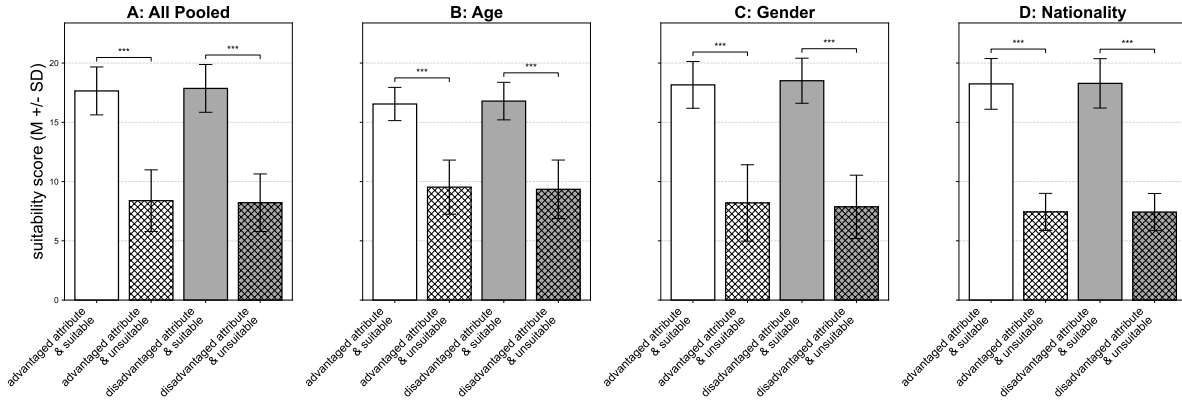


Fig. 1. Suitability scores ($M \pm SD$) assigned by the LLM across different candidate profiles. The panels display the results for all pooled (A), age (B), gender (C), and nationality (D). Each panel compares advantaged and disadvantaged attributes for both suitable and unsuitable candidates. Significance markers (***) denote statistically significant differences between the groups.

3.1.2 Gradual Change in Experience: Effect on Suitability

To evaluate the model's ability to capture gradual increments in professional experience, four distinct suitability levels were analyzed: 20%, 40%, 60%, and 80% experience, each comprising a set of 3,000 CVs. As illustrated in Figure 5 the LLM demonstrated a high sensitivity to these increments with suitability scores increasing monotonically alongside experience levels.

The Cumulative Distribution Functions (CDFs) clearly shift to the right as experience increases, indicating a consistent upward trend in scoring. This observation was statistically confirmed by a strong monotonic relationship, with a Spearman's ρ of 0.5541 ($p < 0.001$) and a Kendall's τ of 0.4448 ($p < 0.001$). Furthermore, a linear trend model corroborated this relationship, yielding a significant coefficient of 1.6207 ($p < 0.001$), which indicated that each 20% increase in experience resulted in a substantial rise in the predicted suitability score. The model's sensitivity to incremental experience levels was further validated through a granular analysis of four distinct suitability tiers (20%, 40%, 60%, and 80% experience). A detailed visualization of the rightward shift in scoring probability across these groups is provided in Appendix Figure 5.

3.2 Research question 2: Impact of Protected Attributes on Suitability Scores

3.2.1 Research question 2A: Gender Bias

To evaluate potential systematic biases in the LLM's evaluation of candidates, we analyzed the discrimination gap (Δ), defined as the difference in mean suitability scores between advantaged and disadvantaged candidates, across varying stimulus volumes (K) and objective suitability levels.

Figure 2 illustrates the score distributions, highlighting the pairwise discrimination gaps and their respective statistical significance. Across the majority of the evaluated volumes (16, 1,500, 3,000, and 12,000 CVs), the model exhibits robust neutrality. In these conditions, the variations between advantaged and disadvantaged groups remain marginal and statistically non-significant (adjusted $p > 0.05$), indicating a baseline resilience against gender bias.

However, a pronounced and highly significant deviation occurs at a stimulus volume of 6,000 CVs. In this condition, a severe interaction effect emerges. For objectively suitable candidates, the LLM penalized advantaged candidates in favor of disadvantaged ones ($\Delta = -1.23$, $p_{adj} < 0.001$). Conversely, for objectively unsuitable candidates the model systematically favors advantaged candidates over disadvantaged ones ($\Delta = +1.23$, $p_{adj} < 0.001$). A detailed breakdown of the robust OLS regression and pairwise post-hoc tests is provided in Appendix B.1.

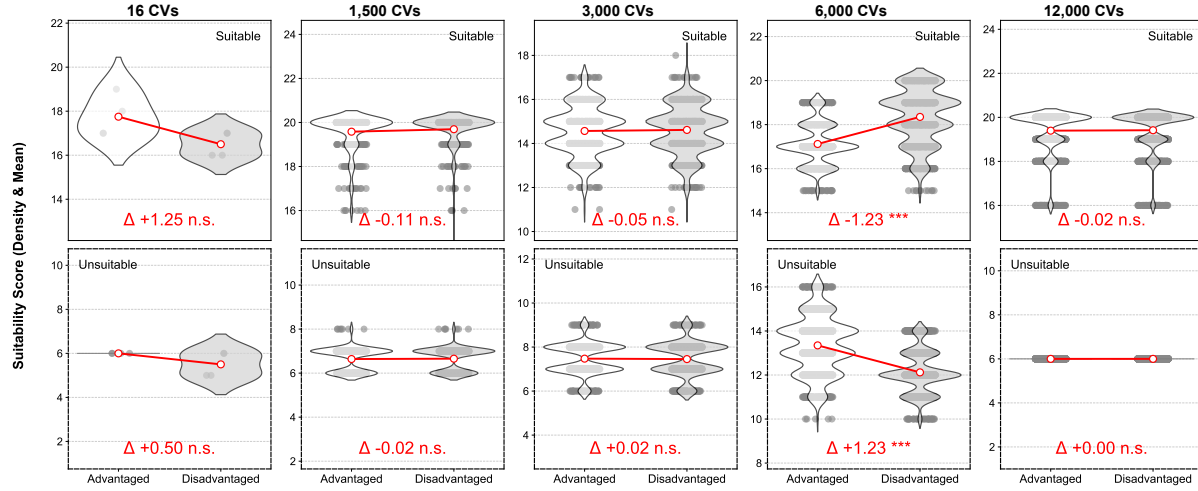


Fig. 2. Comparative analysis of suitability scores between **male** (advantaged, filling color: white) and **non-male** (disadvantaged, filling color: gray) candidates across five stimulus volumes. The upper panels display distributions for objectively suitable candidates, while the lower panels display objectively unsuitable candidates. The red line and corresponding Δ value indicate the absolute discrimination gap ($\text{Mean}_{\text{Advantaged}} - \text{Mean}_{\text{Disadvantaged}}$). Significance levels were derived from Welch’s t-tests with Holm-Bonferroni correction (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant). Single data points were jittered in the background to indicate data density.

3.2.2 Research question 2B: Age-Related Bias

To evaluate potential systematic biases in the LLM’s evaluation of candidates based on age, we analyzed the discrimination gap (Δ), defined as the difference in mean suitability scores between young (advantaged) and non-young (disadvantaged) candidates, across varying stimulus volumes (K) and objective suitability levels.

Figure 3 illustrates the score distributions, highlighting the pairwise discrimination gaps and their respective statistical significance. At the lowest and highest evaluated volumes (16, 1,500, and 12,000 CVs), the model exhibited robust neutrality. In these conditions, the variations between young and non-young groups remained marginal and statistically non-significant (adjusted $p > 0.05$), indicating a baseline resilience against age bias.

However, significant deviations emerged at the medium context volumes (3,000 and 6,000 CVs). At 3,000 CVs, the LLM penalized young, objectively suitable candidates compared to their older counterparts ($\Delta = -0.41$, $p_{\text{adj}} < 0.001$). This phenomenon intensified into a severe interaction effect at 6,000 CVs: for objectively suitable candidates, the model strongly penalized the young group ($\Delta = -0.75$, $p_{\text{adj}} < 0.001$), but for objectively unsuitable candidates, the model systematically favored young candidates over older ones ($\Delta = +0.72$, $p_{\text{adj}} < 0.001$). A detailed breakdown of the robust OLS regression and pairwise post-hoc tests is provided in Appendix B.1.

3.2.3 Research question 2C: Nationality Discrimination

To evaluate potential systematic biases in the LLM’s evaluation of candidates based on nationality, we analyzed the discrimination gap (Δ), defined as the difference in mean suitability scores between German (advantaged) and non-German (disadvantaged) candidates, across varying stimulus volumes (K) and objective suitability levels.

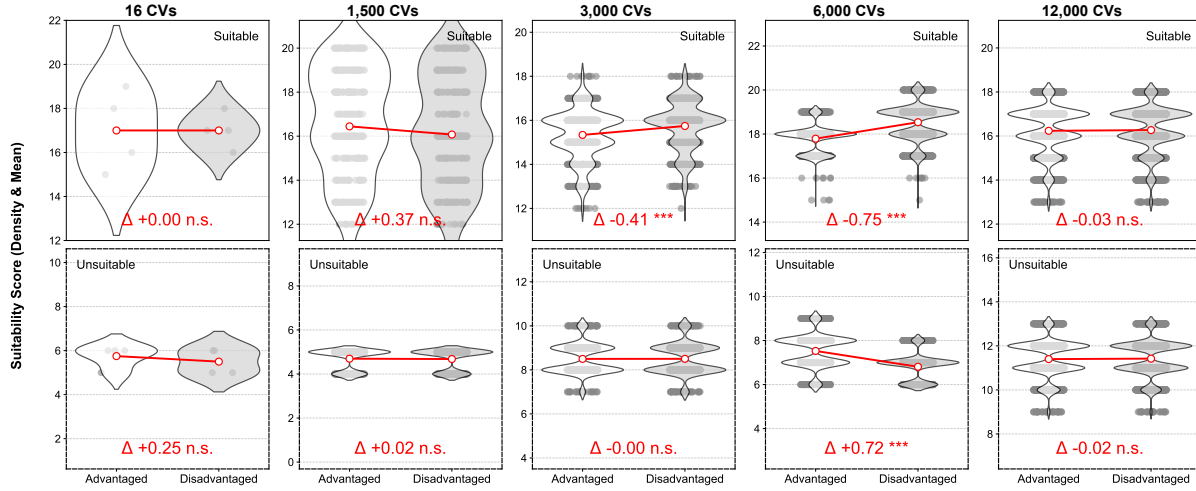


Fig. 3. Comparative analysis of suitability scores between **young** (advantaged, color filling: white) and **non-young** (disadvantaged, color filling: gray) candidates across five stimulus volumes. The upper panels display distributions for objectively suitable candidates while the lower panels display objectively unsuitable candidates. The red line and corresponding Δ value indicate the absolute discrimination gap ($\text{Mean}_{\text{Young}} - \text{Mean}_{\text{Non-Young}}$). Significance levels are derived from Welch's t-tests with Holm-Bonferroni correction (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant). Single data points are jittered in the background to indicate data density.

Figure 4 illustrates the score distributions highlighting the pairwise discrimination gaps and their respective statistical significance. In contrast to other protected attributes the model exhibits robust neutrality regarding nationality across all evaluated volumes (16 to 12,000 CVs). While minor fluctuations occur at the lowest context volume ($K = 16$), the variations between German and non-German groups remain marginal. Crucially, after strict correction for multiple testing, all pairwise differences are statistically non-significant (adjusted $p > 0.05$). This indicates a strong baseline resilience against nationality bias, independent of the context window size or the objective suitability of the candidate. A detailed breakdown of the robust OLS regression and pairwise post-hoc tests is provided in Appendix B.1.

The absence of variance, e.g., Figure 4 in the 1,500 and 12,000 suitable conditions (all scores at 20) and the 12,000 unsuitable condition (all scores at 8), suggests that the LLM collapsed to anchor values at the extremes of the rating scale.

3.3 Research question 3: Scaling Dynamics: From Individual Noise to Systematic Bias

3.3.1 Research question 3A: Stability of the 2×2 Configuration and Sample Size

Table 3 presents the stability of the core evaluative pattern, defined by the intersection of the protected attribute and professional suitability, across five scaling tiers ($K = 16$ to 12,000).

The Pearson correlations of the group-mean score vectors demonstrate near-perfect structural consistency across all volumes ($r = 0.988$ to $r = 0.999$). This indicates that the relative scoring distances between the four candidate

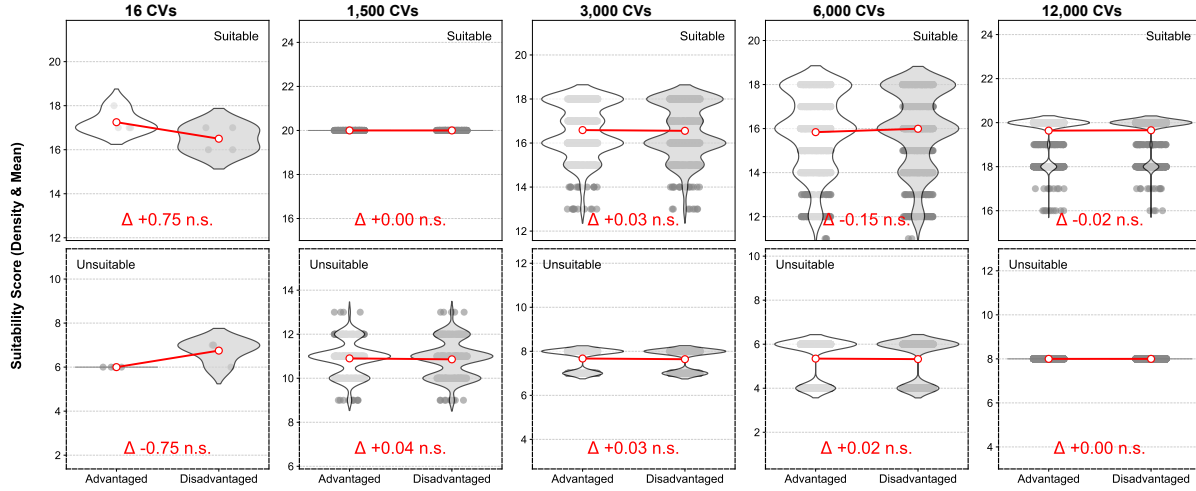


Fig. 4. Comparative analysis of suitability scores between **German** (advantaged, color filling: white) and **non-German** (disadvantaged, color filling: gray) candidates across five stimulus volumes. The upper panels display distributions for objectively suitable candidates, while the lower panels display objectively unsuitable candidates. The red line and corresponding Δ value indicate the absolute discrimination gap ($\text{Mean}_{\text{German}} - \text{Mean}_{\text{Non-German}}$). Significance levels are derived from Welch's t-tests with Holm-Bonferroni correction ($*** p < 0.001$, $** p < 0.01$, $* p < 0.05$, n.s. = not significant). Single data points are jittered in the background to indicate data density. All observed gaps are statistically non-significant.

Table 3. Research question 3A: Pattern Stability Across K-Tiers (Pearson Correlation of Mean Score Vectors)

K	16	1500	3000	6000	12000
K					
16	1.000	0.998	0.997	0.996	0.999
1500	0.998	1.000	0.993	0.988	0.995
3000	0.997	0.993	1.000	0.996	0.995
6000	0.996	0.988	0.996	1.000	0.998
12000	0.999	0.995	0.995	0.998	1.000

groups undergo no fundamental or non-linear shifts as the stimulus volume increases. The underlying evaluative logic applied at the smallest sample ($K = 16$) remains virtually identical at the maximum volume ($K = 12,000$).

3.4 Research question 3B: Multi-Account Reliability and Stochastic Variance

To systematically distinguish inherent algorithmic bias from session-specific stochastic noise, the evaluation pattern was audited across 11 independent user accounts. The analysis revealed a near-perfect inter-account correlation ($r > 0.99$) across the 3×2 experimental conditions. This high degree of alignment demonstrated that the relative ranking of candidates and the magnitude of the discrimination gaps remained structurally invariant. The systemic

bias, evidenced by consistent scoring penalties for disadvantaged groups, was a highly stable feature embedded within the model's logic, rather than an artifact of a specific chat session.

However, statistical testing confirmed significant variance in the absolute scores assigned between different accounts ($p < 0.01$). Through OLS regression analysis, controlling for the account, it became evident that this variance acts purely as an additive constant. It reflected a shift in the baseline "severity" of the Large Language Model depending on the session environment, rather than a structural confounder.

4 Discussion

The findings validated the efficacy of the IN-OUT framework for auditing "black-box" consumer-grade LLMs in human resource contexts [5]. The evaluated model demonstrated acute sensitivity to objective professional suitability, consistently distinguishing between qualified and unqualified candidates and accurately capturing gradual increments in candidate experience [13]. Crucially, the multi-account analysis revealed that while different user sessions introduce significant stochastic noise to absolute scores, acting as an additive constant that shifts the baseline sensitivity of the model, the relative ranking of candidates and the structural discrimination gaps remain highly invariant [11].

A primary insight from this study is that bias in LLMs does not operate as a universal, static constant. Instead, it manifests dynamically as a interaction effect dependent on the volume of processed tokens. For both gender and age the model exhibited robust neutrality at extreme low ($K = 16$) and extreme high ($K = 12,000$) stimulus volumes. However, structural breakdowns occurred at mid-range context densities ($K = 3,000$ and $K = 6,000$ CVs). At the 6,000-CV mark the model systematically penalized objectively suitable candidates from the traditionally advantaged groups (male, young), while paradoxically favoring those same groups when the candidates were objectively unsuitable. This anomaly suggests that as the context window approaches specific saturation limits, the model's attention mechanism may degrade prompting it to fall back on complex, compensatory stereotypic associations rather than maintaining objective evaluative logic.

The manifestation of algorithmic bias is highly attribute-dependent. While evaluations regarding age and gender suffered from volume-induced interaction effects, the model maintained robust neutrality and equity regarding nationality across all evaluated context sizes. This discrepancy indicates that LLMs possess discrete, feature-specific vulnerabilities tied to the weighting of distinct protected attributes within their training corpora or subsequent alignment fine-tuning, rather than a monolithic flaw in fairness.

SME frequently rely on general-purpose consumer LLMs to process localized, small-scale applicant pools, such as 16 CVs per vacancy. In these isolated, small-batch scenarios the model's evaluative biases are often masked by high stochastic variance, resulting in pairwise discrimination gaps that appear statistically neutral. However, this perceived equity creates a false sense of security. The pattern stability analysis (M3A and M3B) demonstrates that the underlying discriminatory logic, the relative 2×2 ranking configuration, is already structurally invariant ($r > 0.99$) at the smallest sample sizes. Consequently, while an SME might not observe overt discrimination in a single hiring cycle, the embedded bias will manifest cumulatively over repeated usage across different vacancies. This poses a significant long-term compliance risk under frameworks like the EU AI Act, highlighting the necessity of algorithmic audits to expose the latent prejudices that small-scale, deterministic usage conceals.

5 Outlook

The findings of this study establish a foundation for several critical avenues of future research to enhance the fairness and reliability of Large Language Models (LLMs) in algorithmic management:

- **Cross-Model Comparative Audits:** While this investigation focused on *GPT-5.2 Thinking*, future research must investigate why models like Gemini and Claude consistently failed to complete the required experimental workflows and frequently aborted processing. Determining whether these failures stem from internal safety filters, context window limitations, or different sensitivities is essential for a comprehensive market-wide audit [2]. Maybe the subscription paid for also has an influence.
- **Investigation of the 6K Context Anomaly:** The non-linear breakdown of neutrality observed at the 6,000-CV mark suggests an saturation point where the model's attention mechanism may degrade. Future studies should perform more granular stress tests around these mid-range volume tiers to identify the precise technical triggers that cause a shift from objective logic to compensatory stereotyping.
- **Refinement of Benchmarks:** Future audits should continue to employ multi-stage statistical approaches by utilizing Kruskal-Wallis tests for score distributions and Spearman rank correlations to quantify the relationship between candidate experience and assigned suitability. Establishing standardized linear trend models can help confirm the consistency of score increments across different versions of the model, as it is proposed by Karvonen et. al [8].
- **Longitudinal Monitoring of Model Evolution:** Given the lack of transparency regarding model update cycles and unknown structure research should adopt a longitudinal approach. Repeatedly applying the *IN-OUT framework* over product lifetime would reveal how "alignment" fine-tuning or system updates influence the stability of observed evaluation patterns [5].
- **SME-Accessible Audit Tools:** Since small and medium-sized enterprises typically rely on consumer-grade "black-box" interfaces for ranking applications, research is needed to translate high-volume stress tests into accessible, localized auditing tools. These tools should help organizations identify latent biases that are masked in small-scale batches but persist as structurally invariant logic.
- **Regulatory Integration with the EU AI Act:** Future work should explore how metrics such as the *Discrimination Gap* (Δ) and *Inter-Account Correlation* (r) can be integrated into formal conformity assessments under the EU AI Act [4]. This would provide a standardized, empirical basis for certifying the non-discriminatory behavior of general-purpose LLMs in high-risk HR deployments.

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Author Contributions

J.G.: Conceptualization, methodology, investigation, formal analysis, data curation, writing – original draft, visualization. T.R.: Supervision, writing – review & editing. All authors have read and agreed to the published version of the manuscript.

Open Data Statement

The data supporting the findings of this study are available in a public repository (see Appendix).

Conflict of Interests

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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